

Pavan Vynatheya

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Research Experience

Canadian Institute for Theoretical Astrophysics, University of Toronto

CITA POSTDOCTORAL FELLOW

Toronto, Canada

Sep. 2024 -

Max Planck Institute for Astrophysics

POSTDOCTORAL RESEARCHER

Garching, Germany

May. 2024 - Aug. 2024

Max Planck Institute for Astrophysics

PHD STUDENT (PIs: ADRIAN S. HAMERS, SELMA DE MINK)

Garching, Germany

Sep. 2020 - Apr. 2024

Inter-University Centre for Astronomy and Astrophysics

MASTER THESIS (PI: KANAK SAHA)

Pune, India

Aug. 2019 - May. 2020

University of Western Ontario

MITACS GLOBALINK RESEARCH FELLOWSHIP (PI: PAULINE BARMBY)

London, Canada

May. 2019 - Aug. 2019

Brera Astronomical Observatory

SUMMER RESEARCH INTERNSHIP (PI: TOMASO BELLONI)

Merate, Italy

May. 2018 - Jul. 2018

Aryabhata Research Institute of Observational Sciences

INDIAN ACADEMY OF SCIENCES SUMMER RESEARCH FELLOWSHIP (PI: SAURABH SHARMA)

Nainital, India

May. 2017 - Jul. 2017

Indian Institute of Astrophysics

SUMMER RESEARCH INTERNSHIP (PI: ANNAPURNI SUBRAMANIAM)

Bengaluru, India

Jun. 2016 - Jul. 2016

Education

Max Planck Institute for Astrophysics

PHD IN ASTRONOMY (DEGREE FROM LUDWIG MAXIMILIAN UNIVERSITY OF MUNICH)

Garching, Germany

Part of the International Max Planck Research School (IMPRS) Munich

Sep. 2020 - May. 2024

Overall grade: *magna cum laude*

Indian Institute of Science Education and Research (IISER) Kolkata

BS-MS DUAL DEGREE IN PHYSICS (DST INSPIRE FELLOW)

Mohanpur, India

Major in Physics + Minor in Mathematics

Cumulative Grade Point Average: 9.19 out of 10

Aug. 2015 - Jul. 2020

Publications

FIRST AUTHOR

- Vynatheya, Pavan; Dessart, Luc; Ryu, Taeho; Pakmor, Rüdiger: 2026, A&A, 709, A214
"Asymmetrical thermonuclear supernovae triggered by the tidal disruption of white dwarfs"
- Vynatheya, Pavan; Ryu, Taeho; Wang, Chen; Sills, Alison; Pakmor, Rüdiger: 2026, ApJ, 999, 64
"The collision and merger products of stars do not look alike: A magnetohydrodynamics comparison"
- Vynatheya, Pavan; Ryu, Taeho; Pakmor, Rüdiger; de Mink, Selma E.; Perets, Hagai B.: 2024, A&A, 685, A45
"Simulating the tidal disruption of stars by stellar-mass black holes using moving-mesh hydrodynamics"
- Vynatheya, Pavan; Mardling, Rosemary A.; Hamers, Adrian S.: 2023, MNRAS, 525, 2388
"Quadruple-star systems are not always nested triples: a machine learning approach to dynamical stability"
- Vynatheya, Pavan; Hamers, Adrian S.; Mardling, Rosemary A.; Bellinger, Earl P.: 2022, MNRAS, 516, 4146
"Algebraic and machine learning approach to hierarchical triple-star stability"
- Vynatheya, Pavan; Hamers, Adrian S.: 2021, ApJ, 926, 195
"How important is secular evolution for black hole and neutron star mergers in 2+2 and 3+1 quadruple-star systems?"

CO-AUTHOR

- Chen, Jing-Qi; Wang, Chen; Ryu, Taeho; Vynatheya, Pavan; Jiang, Dengkai; Moyano, Facundo D.; Chen, Xuefei; Liu, Zhengwei; Han, Zhanwen: 2026 (Submitted to MNRAS)
"Three-dimensional hydrodynamical simulations of main-sequence star mergers: implications for blue straggler formation"

4. Preece, Holly P.; Vigna-Gómez, Alejandro; Rajamuthukumar, Abinaya Swaruba; **Vynatheya, Pavan**; Klencki, Jakub: 2024, arXiv:2412.14022 (Submitted to A&A)
"The Evolution of Massive Stellar Multiplicity in the Field I. Numerical simulations, long-term evolution and final outcome"
3. Britavskiy, Nikolay; Renzo, Mathieu; Nazé, Yael; Rauw, Gregor; **Vynatheya, Pavan**: 2024, A&A, 684, A35
"Tracing the evolution of short-period binaries with super-synchronous fast-rotators"
2. Hamers, Adrian S.; Rantala, Antti; Neunteufel, Patrick; Preece, Holly; **Vynatheya, Pavan**: 2021, MNRAS, 502, 4479
"Multiple Stellar Evolution: a population synthesis algorithm to model the stellar, binary, and dynamical evolution of multiple-star systems"
1. Bogensberger, David; Ponti, Gabriele; Jin, Chichuan; Belloni, Tomaso M.; Pan, Haiwu; Nandra, Kirpal; Russell, Thomas D.; Miller-Jones, James C. A.; Muñoz-Darias, Teo; **Vynatheya, Pavan**; Vincentelli, Federico: 2020, A&A, 641, A101
"An underlying clock in the extreme flip-flop state transitions of the black hole transient Swift J1658.2-4242"

Awards & Fellowships

2024-27	CITA Postdoctoral Fellowship , Postdoctoral research at the Canadian Institute for Theoretical Astrophysics, University of Toronto	Toronto, Canada
2023	Poster Award , 3,2,1: Massive Triples, Binaries and Mergers conference – poster on the dynamical stability of triple- and quadruple-star systems	Leuven, Belgium
2022	Kippenhahn Award , Best PhD student paper in 2021 at the Max Planck Institute for Astrophysics – paper on gravitational wave progenitors in quadruple-star systems	Garching, Germany
2018	Mitacs Globalink Research Internship , Summer research at the University of Western Ontario	London, Canada
2017	Indian Academy of Sciences Summer Research Fellowship , Summer research at the Aryabhata Research Institute of Observational Sciences	Nainital, India
2015-20	Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship , Awarded by the Department of Science and Technology to study science at IISER Kolkata for 5 years	Mohanpur, India

Talks & Workshops

INVITED TALKS

Sep. 2026	Annual Meeting of the German Astronomical Society , Max Planck Institute for Astrophysics	Garching, Germany
Sep. 2025	CITA seminar , Canadian Institute for Theoretical Astrophysics, University of Toronto	Toronto, Canada
Jun. 2025	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Dec. 2024	TASTY seminar , Department of Astronomy and Astrophysics, University of Toronto	Toronto, Canada
Apr. 2024	MPA institute seminar , Max Planck Institute for Astrophysics	Garching, Germany
Mar. 2024	Collisional N-body dynamics workshop , Max Planck Institute for Astrophysics	Garching, Germany
Oct. 2022	Galaxies, Stars & Cosmology group seminar , Marseille Astrophysics Laboratory	Marseille, France (online)

SEMINAR TALKS

Mar. 2026	Astro seminar , Kavli Insitute for the Physics and Mathematics of the Universe	Kashiwa, Japan
Dec. 2025	Astrophysics & Cosmology seminar , Indian Institute of Science	Bengaluru, India
Oct. 2024	Astronomy journal club , Department of Physics and Astronomy, McMaster University	Hamilton, Canada
Oct. 2024	CITA Jamboree , Canadian Institute for Theoretical Astrophysics, University of Toronto	Toronto, Canada
Sep. 2023	Stars and Compact Objects group seminar , Center for Computational Astrophysics, Flatiron Institute	New York, USA
Aug. 2023	Theoretical Astrophysics Including Relativity group seminar , Theoretical Astrophysics Including Relativity and Cosmology, California Institute of Technology	Pasadena, USA
Aug. 2023	Stellar & Galactic Astronomy group seminar , Division of Astronomy & Astrophysics, University of California Los Angeles	Los Angeles, USA
Jun. 2023	Massive stars and stellar populations group seminar , Anton Pannekoek Institute for Astronomy, University of Amsterdam	Amsterdam, Netherlands
Apr. 2023	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Nov. 2022	Astrophysical relativity group seminar , International Centre for Theoretical Sciences	Bengaluru, India
Oct. 2022	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Sep. 2022	Astrophysics whiteboard session , School of Physics & Astronomy, Monash University	Clayton, Australia
Oct. 2021	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Jul. 2021	Max Planck Institute for Astrophysics summer retreat , Schloss Ringberg	Kreuth, Germany

CONTRIBUTED TALKS

May. 2026	IAU Symposium 406: Future Landscape of Astrophysical Transients , Hotel Kakola (organized by University of Turku)	Turku, Finland
Feb. 2026	Multi-Messenger Astrophysics in the Dynamic Universe , Yukawa Institute for Theoretical Physics, Kyoto University	Kyoto, Japan
Jun. 2025	IAU Symposium 398 & MODEST-25: Compact Objects and Binaries in Dense Stellar Systems , Seoul National University	Seoul, South Korea
Apr. 2024	Anticipating the Rising Tide of Tidal Disruption Events: Theory and Observations , Kavli Institute for Theoretical Physics, University of California Santa Barbara	Santa Barbara, USA
Sep. 2023	“Two in a million” – The interplay between binaries and star clusters , European Southern Observatory	Garching, Germany
Aug. 2023	MODEST-23: Star clusters in the post-pandemic era , Northwestern University	Evanston, USA
May. 2022	Growing Black Holes: Accretion and Mergers , Radisson Hotel Kathmandu (organized by Astrophysics and Space Science Observatory of Bologna)	Kathmandu, Nepal
Jan. 2022	Dynamical Formation of Gravitational Wave Sources , Aspen Center for Physics	Aspen, USA

SCHOOLS

Jan. 2024	Nordic Winter School on Multimessenger Astrophysics , Thon Hotel Skeikampen	Skeikampen, Norway
Jul. 2022	Summer School for Astrostatistics at Crete , Department of Physics, University of Crete	Heraklion, Greece
Aug. 2021	NBIA Summer School on Gravitational Wave Astrophysics , Niels Bohr International Academy, University of Copenhagen	Copenhagen, Denmark
Dec. 2018	IIST Astronomy and Astrophysics School , Indian Institute of Space Science and Technology	Thiruvananthapuram, India
Feb. 2017	Telescope-making workshop , Indian Institute of Science Education and Research Kolkata	Mohanpur, India
Dec. 2015	Vijyoshi Science Camp , Eastern Zonal Cultural Centre (organized by Indian Institute of Science Education and Research Kolkata)	Kolkata, India

CONFERENCE POSTERS

Jul. 2023	3,2,1: Massive Triples, Binaries and Mergers , Katholieke Universiteit Leuven	Leuven, Belgium
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Students

2024-	Gurman Sachdeva , Undergrad/grad student	Toronto, Canada
2025-	Yung-Hsin (Stephanie) Lee , Post-undergrad student	Toronto, Canada
2026-	Estella Irvine , Summer intern	Toronto, Canada
2026-	Shuo (Michael) Chen , Summer intern	Toronto, Canada

Skills

Astrophysics code	Moving-mesh hydrodynamics: AREPO , Multiple-star population synthesis: MSE , <i>N</i> -body: MSTAR , REBOUND , Detailed stellar evolution: MESA
Programming	python (advanced + machine learning), C/C++ (intermediate), Java (basic)
Web design	HTML (advanced), CSS (advanced), jQuery/Javascript (intermediate), PHP (basic)
Languages	English (fluent), Kannada (native), Tamil (native), Hindi (fluent)

Activities

2025-	Referee , Astrophysical Journal (ApJ), Publications of the Astronomical Society of Japan (PASJ)	
2021-23	Student representative , International Max Planck Research School (IMPRS) Munich	Garching, Germany
2021-22	Co-organizer , Seminar on Stellar Astrophysics (SESTAS)	Garching, Germany
2018 and 2019	Web designer , IISER Kolkata team at the International Genetically Engineered Machine (iGEM) competition - Gold medal winners	Boston, USA (did not attend)
2016-17	Web designer , Inquivesta - Science fest of IISER Kolkata	Mohanpur, India
2016-17	Convener , IISER Kolkata Music club	Mohanpur, India